

Lec: 16.

ROTATION & UNITARY MATRICES

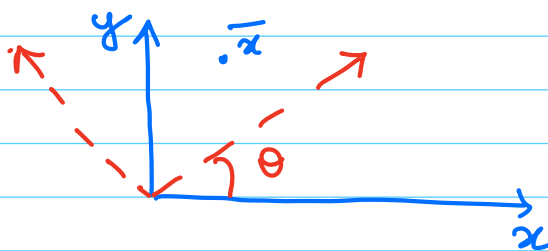
classes of matrices

Rotation matrix

$R(\theta)$ $\xrightarrow{2 \times 2}$ $\xrightarrow{\text{Parameter}}$

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$R(\theta)$ corresponds to rotation of coordinate system anti clock wise by θ .



Note: R satisfies the property

$$R^T R = R R^T = I$$

$$\boxed{R^T = R^{-1}}$$

$$\begin{aligned} R^T R &= \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2 \times 2} \end{aligned}$$

Similarly, $R R^T = I$.

UNITARY MATRIX
 "complex Rotation matrix"

unitary matrix $U \rightarrow U U^H = U^H U = I$

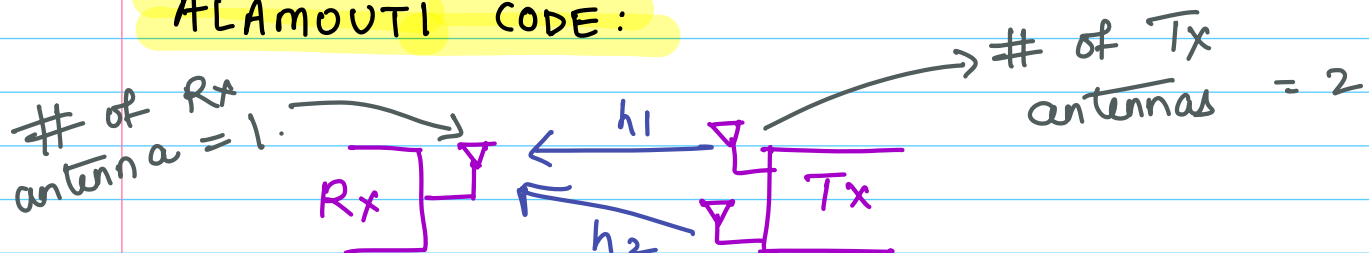
example: $U = \begin{bmatrix} e^{j\phi_1} \cos \theta & e^{j\phi_2} \sin \theta \\ -e^{-j\phi_2} \sin \theta & e^{-j\phi_1} \cos \theta \end{bmatrix}$
 3 parameters ϕ_1, ϕ_2 & θ .

$$U U^H = \begin{bmatrix} e^{j\phi_1} \cos \theta & e^{j\phi_2} \sin \theta \\ -e^{-j\phi_2} \sin \theta & e^{-j\phi_1} \cos \theta \end{bmatrix} \begin{bmatrix} e^{-j\phi_1} \cos \theta & -e^{j\phi_2} \sin \theta \\ e^{-j\phi_2} \sin \theta & e^{j\phi_1} \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -e^{j(\phi_1+\phi_2)} \cos \theta \sin \theta + e^{j(\phi_1+\phi_2)} \cos \theta \sin \theta \\ -e^{-j(\phi_1+\phi_2)} \cos \theta \sin \theta + e^{-j(\phi_1+\phi_2)} \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

ALAMOUTI CODE:



h_1, h_2 = channel coefficients.

H ← effective channel matrix for Alamouti code

$$= \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix}$$

unitary matrix

$$\tilde{H} = \frac{1}{\|\bar{h}\|} \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix}$$

→ $\|\bar{h}\| = \sqrt{|h_1|^2 + |h_2|^2}$

$$\tilde{H}^H \tilde{H} = \tilde{H} \tilde{H}^H = \mathbf{I}_{2 \times 2}$$

Properties of Unitary matrices.

$$|\det(U)| = 1.$$

$$UU^H = \mathbf{I}$$

$$\Rightarrow \det(UU^H) = \det(\mathbf{I}) = 1$$

$$\Rightarrow \det(U) \cdot \det(U^H) = 1$$

$$\Rightarrow \det(U) \cdot (\det(U))^* = 1.$$

$$\Rightarrow |\det(U)|^2 = 1$$

$$\Rightarrow |\det(U)| = 1.$$

Let λ be eigenvalue of U
then $|\lambda| = 1.$

Let $\bar{z}$ = corresponding eigenvector

$$\Rightarrow U \bar{z} = \lambda \bar{z} \quad (1)$$

$$\Rightarrow \bar{z}^H U^H = \lambda^* \bar{z}^H \quad (2)$$

Multiplying LHS and RHS

in (1), (2).

$$\Rightarrow \bar{z}^H \underbrace{U^H U}_{I} \bar{z} = \lambda^* \bar{z}^H \lambda \bar{z}$$

$$\Rightarrow \bar{z}^H \bar{z} = |\lambda|^2 \|\bar{z}\|^2$$

$$\Rightarrow \|\bar{z}\|^2 = |\lambda|^2 \|\bar{z}\|^2$$

$$\Rightarrow |\lambda| = 1$$

→ eigenvalues of Unitary matrix have 'unit' magnitude!

$$U^H U = I$$

$$\Rightarrow \begin{bmatrix} \bar{u}_1^H \\ \bar{u}_2^H \\ \vdots \\ \bar{u}_n^H \end{bmatrix} \begin{bmatrix} \bar{u}_1 & \bar{u}_2 & \dots & \bar{u}_n \end{bmatrix}$$

non-diagonal entries = 0

Diagonal entries = 1

$$= \begin{bmatrix} \bar{u}_1^H \bar{u}_1 & \bar{u}_1^H \bar{u}_2 & \dots & \bar{u}_1^H \bar{u}_n \\ \bar{u}_2^H \bar{u}_1 & \bar{u}_2^H \bar{u}_2 & \dots & \bar{u}_2^H \bar{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \bar{u}_n^H \bar{u}_1 & \bar{u}_n^H \bar{u}_2 & \dots & \bar{u}_n^H \bar{u}_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{bmatrix}$$

$$\Rightarrow \bar{u}_i^H \bar{u}_i = \|\bar{u}_i\|^2 = 1$$

$$\bar{u}_i^H \bar{u}_j = 0 \quad \text{if } i \neq j.$$

columns $\bar{u}_i$ are unit norm & orthogonal

$\Rightarrow$ orthonormal

Unitary matrix has orthonormal columns.

Similarly rows are also orthonormal.

$$\tilde{x} \xleftarrow{n \times 1} = U \bar{x} \xrightarrow{n \times 1}$$

$$\begin{aligned} \|\tilde{x}\|^2 &= \tilde{x}^H \tilde{x} \\ &= \bar{x}^H \underbrace{U^H U}_{\mathbf{I}} \bar{x} \\ &= \bar{x}^H \bar{x} = \|\bar{x}\|^2 \end{aligned}$$

$$\Rightarrow \|\tilde{x}\| = \|\bar{x}\|$$

$$\tilde{x} = U \bar{x}$$

$\hookrightarrow$ multiplication w/ unitary matrix leaves norm invariant.

i.i.d. Gaussian R.V.

$$\bar{x} \sim N(0, \sigma^2 \mathbf{I})$$

$$E\{\bar{x}\} = 0$$

$$E\{\bar{x} \bar{x}^H\} = \sigma^2 \mathbf{I}$$

$$\tilde{x} = U \bar{x}$$

$$\begin{aligned} E\{\tilde{x}\} &= E\{U \bar{x}\} \\ &= U E\{\bar{x}\} \\ &= U \times 0 = 0. \end{aligned}$$

$$\begin{aligned} E\{\tilde{x} \tilde{x}^H\} &= E\{U \bar{x} \bar{x}^H U^H\} \\ &= U E\{\bar{x} \bar{x}^H\} U^H \\ &= U \sigma^2 \mathbf{I} U^H \\ &= \sigma^2 U U^H = \sigma^2 \mathbf{I}. \end{aligned}$$

$$\therefore E\{\tilde{x}\} = 0$$

$$E\{\tilde{x} \tilde{x}^H\} = \sigma^2 \mathbf{I}$$

multi-variate
Gaussian

$$\tilde{x} \sim N(0, \sigma^2 \mathbf{I})$$

→ i.i.d Gaussian elements
mean = 0 variance = $\sigma^2 \mathbf{I}$.

